
CYBER THREAT MONITORING LEVEL I

Day 2: Solving Routine Problems on a Real Desk

Student Handout

Today's Units

Code	Unit of Competency
400311103	Solve/address routine problems
400311104	Enhance self-management skills
400311105	Support innovation
400311109	Adopt entrepreneurial mindset in the workplace

Problem vs Symptom

What people report (symptom)	What is actually wrong (problem)
"The dashboard is empty"	The agent stopped sending events
"I'm getting a hundred alerts"	One rule is matching normal activity
"The same ticket keeps appearing"	The integration is retrying on timeout
"The scan never finished"	The target went offline mid-scan
"Antivirus cleaned it but it's back"	The persistence mechanism was not removed

If you fix the symptom, the problem comes back tomorrow.

Three Questions That Turn a Symptom Into a Problem Statement

1. **What exactly is happening?** — observable and specific, with no interpretation
2. **When did it start?** — and what else changed at that time
3. **What is the scope?** — one host, one team, one site, or everything

Bad	"The SIEM is broken."
Good	"Since approximately 14:20 today, four Manila servers (SRV-MNL-01 to 04) have shown no events in the dashboard. All other hosts are reporting normally. The four affected hosts respond to ping."

A good problem statement contains a **start time**, a **scope**, a **boundary** (what is *not* affected), and at least one piece of **evidence**.

The Five Whys

Ask "why" until the answer stops being a symptom and becomes something you can change.

PROBLEM: The same malware detection reappears on WKS-118 every morning.

Why? → Antivirus detects and cleans the file each morning.

Why does it come back? → Something writes it back to disk.

Why? → A scheduled task runs at logon and downloads it.

Why is there a scheduled task? → It was created by the original infection.

Why wasn't it removed? → AV removed the FILE but not the PERSISTENCE.

ROOT CAUSE: Remediation was incomplete.

ACTION: Remove the scheduled task, re-scan, verify at next logon.

Two Rules for the Five Whys

- **Stop at the first thing you can act on.** Going further gives you a philosophy, not a fix.
 - **If your answer is a person's name, back up.** "The user clicked a link" is a training gap, not tonight's root cause. Find the technical cause first.
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Cause-and-Effect Analysis (Fishbone)

For any SOC fault, check these six branches:

Branch	Ask
Agent	Is the service running? Right version? Right config?
Network	Can the host reach the server? Is a firewall rule blocking it?
Server	Is the manager up? Is the disk full? Is it accepting writes?
Configuration	Did somebody change a rule, a filter, or a log level?
Host	Did it reboot? Is it patched? Is it even powered on?
Process	Is there a change notice? Did somebody do this deliberately?

Then ask the question that saves your night: "**Which of these can I test in under five minutes?**" Test those first.

Choosing a Corrective Action

Write down **three** options before you choose one. Then apply the filters **in this order**:

#	Filter	If the answer is no
1	Is it within my authority ?	It is not an option — it is an escalation
2	Is it reversible ?	Prefer an option you can undo
3	Is it the smallest thing that could work ?	Restart the service before you reboot the host

The dangerous analyst is not the one who does not know the answer. It is the one who does something big and irreversible because it felt decisive.

The Action Plan

Field	Content
Problem	One paragraph: what, when, scope, boundary
Cause	The root cause, stated as something changeable
Action	Numbered steps — each one a single verifiable act
Who	A named person or role for each step
When	A time or deadline for each step
How we will know it worked	The observable test, with a number
If it does not work	The fallback, and who to escalate to

Define success in advance, with a number. "No detection for three consecutive working days" is testable. "Seems fixed" is not.

Saying the Same Plan Three Ways

Audience	What they need	Length
Your L2 / supervisor	Technical detail, what you already tested, what you need from them	3–4 sentences
The client / user	What is happening, what it means for them, when they hear next	2 sentences, no jargon
The ticket	Everything, in order, with timestamps — the permanent record	As long as it needs

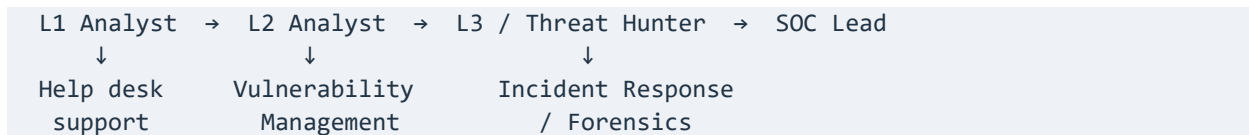
You get about **sixty seconds** with your L2. They have eleven other things happening.

Unit 400311104 — Enhance Self-Management Skills

Personal Goals vs Career Goals

	Personal goal	Career goal
About	Who you want to be	What you want to do
Timeframe	Often lifelong	Usually 1–5 years
Example	"Support my family without being absent from it"	"Move from L1 to L2 within two years"

The Ladder You Are Standing On



MY GOAL SHEET

Name: _____ Date: ____

My personal goal:

My career goal — 1 year from now:

My career goal — 3 years from now:

Two things I must learn to reach the 1-year goal:

1. _____

2. _____

One thing I will do in the next 30 days:

Recognising Emotions on the Desk

State	What it feels like	What it does to your work	What to do
Alert fatigue	Numb, clicking through	You close things without reading	Stand up. Take five. Re-read the last three.

Panic	Racing, tunnel vision	You skip steps and forget to record	Go back to the SOP. Follow it literally.
Frustration	Snappy, blaming the tool	You stop asking for help	Say out loud that you are stuck.
Overconfidence	Certain, fast, skipping checks	You escalate the wrong thing	Verify one thing you "already know".

Positive state	Why it helps	How to protect it
Curiosity	It is the engine of good triage	Give yourself ten minutes on the odd alert
Calm	Lets you follow process under pressure	Comes from having run the drill before
Satisfaction	Sustains you through quiet shifts	Notice the catches, not just the misses

My Reflection

A time I felt frustrated at work or in study:

A time I felt panic or pressure:

A time I felt real satisfaction in my work:

Learning Style Inventory

For each pair, tick the one that is *more* true of you.

#	A	B
1	☐ I learn by trying it myself	☐ I learn by watching first
2	☐ I want the reason before the steps	☐ I want the steps, the reason can come later
3	☐ I like timed drills	☐ I like time to think it through
4	☐ I remember what I did	☐ I remember what I read
5	☐ I ask "what happens if I break it?"	☐ I ask "what is the correct procedure?"
6	☐ I get bored in long explanations	☐ I get anxious when rushed

Mostly A → you learn by doing and experimenting. Mostly B → you learn by watching and thinking it through.

If you learn mainly by...	Easy for you	Hard for you	Your strategy
Doing	Days 6–14	Days 1–3	Ask for a demonstration of every concept
Watching	Demonstrations	Going first	Take notes, then do it immediately while fresh
Thinking it through	SOPs, frameworks	Fast timed drills	Read ahead the night before
Trying and adjusting	Timed drills	Long procedures	Ask "what breaks if I get this wrong?"

My dominant style: _____

The part of this course I expect to find hardest:

One concrete strategy I will use to get through it:

Unit 400311105 — Support Innovation

The Four Signals That Innovation Is Needed

Signal	What it sounds like	Example
Repetition	"I do this every single shift"	Manually copying alert details into the ticket
Workaround	"You just have to know to do X first"	An undocumented step everyone learned the hard way
Complaint	"This always happens on Mondays"	A rule that fires on a scheduled backup job
Near miss	"We nearly missed that one"	A real alert buried under false positives

Ranking an Idea

Criterion	Question
Impact	How much time or risk does it remove, per week?
Effort	One person in a shift, or a project?
Risk	What breaks if it goes wrong? Can we undo it?
Authority	Can an L1 do it, or does it need approval?

SOP IMPROVEMENT PROPOSAL

Proposed by: _____ Date: ____

What I noticed (*repetition / workaround / complaint / near miss*)

Why it matters (*time cost per week, or the risk it creates*)

What I propose

Who needs to approve it

How we would know it worked

Peer review — completed by: _____

One thing I like about this proposal:

One question I have about it:

Unit 400311109 — Entrepreneurial Mindset

What Your Decisions Cost

Thing	What it costs the business
An alert triaged well the first time	~8 minutes of L1 time
The same alert escalated wrongly to L2	~45 minutes of L2 time, plus your time
A missed critical alert	A client incident, a penalty, sometimes the contract
A false escalation at 3 AM	A manager's night — and their trust in your judgement

Nobody hands an L1 a budget. But every triage decision spends somebody's money. Analysts who understand that get promoted, because they escalate the right things.

Entrepreneurial Practices in a SOC

- **Quality assurance** — check your own ticket before you submit it
 - **Cost of rework** — a badly written ticket gets done twice
 - **Reliable sources** — authoritative intel, not a forum post
 - **Customer focus** — the client wants to know what it means for *them*
 - **Reputation** — one wrongly closed critical alert is remembered for years
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Notes
